JAVA MINI PROJECT

Retail Shop Inventory & Product Billing Management System

A simple console-based application for product tracking, stock control and automatic bill calculation.

PROJECT PRESENTATION

Presented by: _____

INVENTORY

03

BILLING

₹130.00

STOCK STATUS

UPDATED

Project overview

THE IDEA

Bring everyday shop operations into one small Java program.

- Keep product IDs, prices and stock quantities together.
- Create bills by selecting products and entering quantities.
- Update stock automatically after a valid sale.

AT A GLANCE

03	Sample products
05	Menu options
₹	Automatic totals

Problem statement & objectives

CHALLENGES

- Manual stock records are slow to maintain.
- Billing calculations can contain mistakes.
- Stock may not be updated after a sale.
- Staff need a quick way to check availability.

PROJECT OBJECTIVES

- Display and maintain product inventory.
- Add products and restock existing items.
- Calculate each bill and the grand total.
- Validate product IDs and quantities.
- Reduce stock after a successful sale.

Technology stack

JAVA

Core programming language

SCANNER

Reads keyboard input

LINKEDHASHMAP

Stores products by ID

CONSOLE UI

Menu-driven interface

Java concepts demonstrated: classes, objects, constructors, methods, loops, conditions and switch.

System modules

01	**View inventory**	See product ID, name, price and stock.
02	**Add product**	Register a product with a unique ID.
03	**Restock**	Increase stock for an existing product.
04	**Generate bill**	Select items and quantities for a sale.
05	**Exit**	Close the application safely.

Workflow & billing logic

1

Start

Load sample products

→

2

Choose

Select a menu option

→

3

Validate

Check ID, quantity and stock

→

4

Calculate

Price × quantity

→

5

Update

Reduce stock and show total

Billing rule Line amount = unit price × quantity • Grand total = sum of line amounts

Sample inventory & bill

ID	Product	Unit price	Available stock
101	Rice	₹50.00	10
102	Sugar	₹40.00	10
103	Milk	₹30.00	10

EXAMPLE CUSTOMER BILL

2 × Rice	**₹100.00**
1 × Milk	**₹30.00**
TOTAL	**₹130.00**

Stock after sale: Rice = 8 units • Milk = 9 units

Program structure

RetailShopInventoryBilling

main()

Menu loop + user choice

Product

name • price • stock

KEY METHODS

showProducts()	Prints current inventory
addProduct()	Adds a new product
restockProduct()	Increases stock quantity
createBill()	Checks stock, totals bill and updates inventory

Limitations & future enhancements

CURRENT LIMITATIONS

- Data is stored in memory only.
- Information resets when the program closes.
- No database, login or graphical interface.
- No tax, discount or printed receipt feature.

FUTURE ENHANCEMENTS

- Store data in MySQL or files.
- Add GST, discounts and invoice numbers.
- Print or export receipts and sales reports.
- Build a GUI and user authentication.
- Add search, low-stock alerts and date/time.

CONCLUSION

A small program.
A practical solution.

The project demonstrates how Java can support basic inventory control,
quantity validation and reliable bill calculation.

THANK YOU

Questions & Discussion

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